(iii) ASA (Angle-Side-Angle) Congruence

If two angles and the included side of a triangle are congruent to two angles and the included side of another triangle, then the triangles are congruent. Here, $\angle A = \angle R$, CA=PR and $\angle C = \angle P$ and hence $\Delta ABC \equiv \Delta RQP$

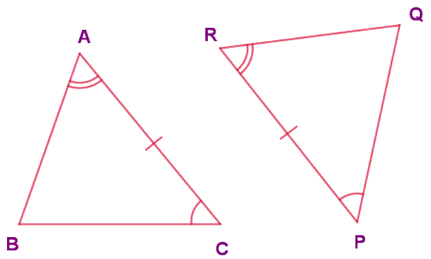


Fig. 5.5

(iv) RHS (Right Angle – Hypotenuse – Side) Congruence

If the hypotenuse and a leg of one right triangle are congruent to the hypotenuse and a leg of another right triangle, then the triangles are congruent. Here, $\angle B = \angle Q = 90°$, $BC = QR$ and $AC = PR$

(right angle) (leg) (hypotenuse)

and hence $\Delta ABC \equiv \Delta PQR$.

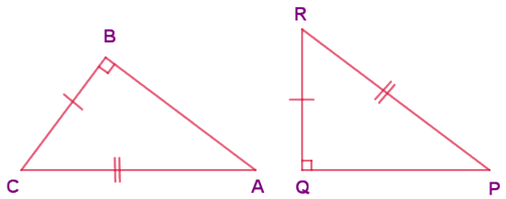


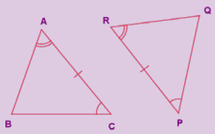
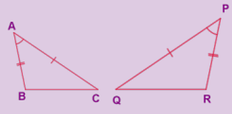
Fig. 5.6

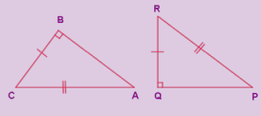
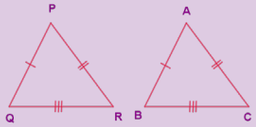
Note

- Any segment or angle is congruent to itself! This is called Reflexive property
- If two triangles are congruent, then their corresponding parts are congruent. This is called **CPCTC (Corresponding parts of Congruent Triangles are Congruent).**
- *If angles then sides* means if two angles are equal in a triangle, then the sides opposite to them are equal.
- *If sides then angles* means if two sides are equal in a triangle, then the angles opposite to them are equal.

Try these

Match the following by their congruence property

S.No.	A	B
1		RHS
2		SSS

S.No.	A	B
3		SAS
4		ASA

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